Single-Phase Power Transformer Sizing

A Practical Methodology for High-Frequency, High-Power Designs

1 Scope

This document summarizes the engineering steps required to dimension a single-phase power transformer for switch-mode converter applications (e.g. PSFB, DAB, isolated DC/DC stages), covering:

- turns ratio determination from the voltage transfer function,
- core material selection as a function of operating frequency,
- core sizing via the area-product method,
- winding design considerations (skin/proximity effects, litz wire vs. foil),
- a worked numerical example at 800 V/800 V, 10 kHz, 250 kW.

2 Turns Ratio

For an ideal isolated transformer driven by a quasi-square or square voltage waveform (as typical of PSFB, DAB, or any hard/soft-switched bridge), the volt-second balance on each winding gives Faraday's law in its waveform-corrected form:

$$V = K_f \, N \, A_c \, B_m \, f \tag{1}$$

where:

- V = RMS (or peak, depending on K_f convention) voltage applied to the winding [V],
- N = number of turns,
- A_c = core cross-sectional area [m^2],
- B_m = peak flux density [T],
- f = switching/operating frequency [Hz],
- K_f = waveform factor: $K_f = 4$ for a symmetric square wave, $K_f = 4.44$ for a pure sinusoid, intermediate values for trapezoidal/quasi-square waveforms with duty cycle $D < 0.5$ (e.g. $K_f \approx 4D$ for a unipolar square wave of duty D).

The turns ratio is fixed by the required voltage transformation:

$$n = \frac{N_1}{N_2} = \frac{V_1}{V_2} \tag{2}$$

For converters requiring voltage step change under varying duty cycle (e.g. PSFB with output inductor), n is chosen at the nominal/maximum duty point, with margin for line and load regulation.

3 Core Material Selection vs. Frequency

The dominant constraint at a given frequency is core loss, which is approximately:

$$P_{core} \approx k_c \, f^{\alpha} \, B_m^{\beta} \, V_{core} \tag{3}$$

(Steinmetz equation), where α, β and k_c are material-dependent. The choice of material is therefore frequency-driven:

Material	Typical f range	B_{sat} [T]	Notes
Grain-oriented Si-Fe (laminated)	DC–1 kHz	1.6–2.0	High saturation, but eddy-current losses scale with f^2t^2; lamination thickness limits practical use above ∼1 kHz.
Amorphous (Fe-based)	1–10 kHz	1.5–1.6	Lower loss than Si-Fe at the same f, good compromise for medium-frequency, high-power transformers.
Nanocrystalline (e.g. Vitroperm, Finemet)	1–150 kHz	1.2–1.5	High B_{sat} retained at MF, low specific losses; preferred for high-power MF transformers (DAB, solid-state transformers, traction).
MnZn ferrite	20 kHz–500 kHz	0.35–0.5	Low B_{sat} limits power density at low kHz; becomes attractive above ∼50 kHz where its losses remain low and nanocrystalline/amorphous losses rise.
NiZn ferrite	>500 kHz	0.3–0.4	High-frequency, low-power applications; not relevant for high-power MF transformers.

Rule of thumb: for a given frequency, select the material that allows the highest usable B_m (limited by core loss density, typically targeting 0.5 W/cm^3 to 1.5 W/cm^3 depending on cooling) while satisfying thermal constraints. At 10 kHz and high power, nanocrystalline or amorphous tape-wound cores typically outperform ferrite because their higher B_{sat} allows a smaller, lighter core for the same volt-seconds, even though their specific core losses per unit volume at 10 kHz are higher than ferrite's — the reduction in required volume usually wins for power levels in the hundreds of kW.

4 Core Sizing: Area-Product Method

The area product $A_p = A_c \cdot A_w$ (core area times window area) relates the apparent power transferred to the achievable current density and flux density, and is the standard first-pass sizing tool (McLyman):

$$A_p \, [\mathrm{cm}^4] = \frac{P_t \cdot 10^4}{K_f \, K_u \, B_m \, f \, J} \tag{4}$$

where:

- P_t = total apparent power handled by the transformer, $P_t \approx P_{in} + P_{out} \approx 2P$ for an isolated transformer carrying power P on each side,

- K_u = window utilization factor ($\approx$ 0.2–0.3 for round/litz wire with insulation and creepage, up to $\approx$ 0.4–0.5 for foil/busbar windings),
- B_m = peak flux density [T] (material- and loss-limited, see above),
- f = frequency [Hz],
- J = RMS current density [A/cm^2] (cooling-dependent: $\sim$200–350 for natural convection, up to 500+ with forced air/liquid cooling).

Once A_p is known, a core (or stack of cores) is selected from a manufacturer catalogue with matching or slightly larger A_p, giving A_c and A_w directly. Equation (1) then yields the turns count for each winding.

5 Winding Design

At MF/HF, conduction losses are dominated by skin and proximity effects rather than DC resistance:

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0 \mu_r}} \tag{5}$$

For copper, $\delta \approx 7.5/\sqrt{f\,[\mathrm{Hz}]}$ cm. At 10 kHz, $\delta \approx 0.66$ mm, meaning conductor thickness should not exceed roughly $2\delta \approx 1.3$ mm in any single layer.

- **Litz wire**: appropriate for moderate current windings where many strands keep each strand below δ; strand count and bundle geometry must be checked for proximity losses in multi-layer windings.
- **Copper foil**: preferred at very high current, low turn-count windings (as in the example below); foil thickness $\leq 2\delta$, interleaved with thin insulation, often the only practical option above a few hundred amperes RMS per turn.
- **Interleaving**: primary/secondary interleaving reduces the MMF excursion across the window and is essential to control proximity losses at MF in high-current designs.

6 Insulation and Isolation

Creepage and clearance distances are dictated by the working voltage, altitude, pollution degree, and required isolation class (basic/supplementary/reinforced) per IEC 60664-1 or the relevant converter standard. Tape-wound cores with bobbin or potted construction are common; for busbar/foil windings, solid insulation (e.g. polyimide, Nomex, epoxy-glass) must be verified for partial-discharge-free operation at the working dV/dt, which can be severe with hard-switched square waves.

7 Worked Example: 800 V/800 V, 10 kHz, 250 kW

Assumptions

- Square-wave excitation (PSFB/DAB-type bridge), $K_f = 4$.
- Turns ratio $n = 1$ (800 V/800 V).
- Nanocrystalline tape-wound core, $B_m = 1.0$ T (derated from $B_{sat} \approx 1.2$ T for MF loss margin).
- $K_u = 0.3$ (foil winding, generous insulation/creepage at this power level).
- $J = 300$ A/cm^2 (forced-air or liquid-cooled busbar/foil winding).

Step 1 — Apparent power

$$P_t = 2P = 2 \times 250\,\text{kW} = 500\,\text{kW} \tag{6}$$

Step 2 — Area product

$$A_p = \frac{500{,}000 \times 10^4}{4 \times 0.3 \times 1.0 \times 10{,}000 \times 300} \approx 1389\,\text{cm}^4 \tag{7}$$

This is a large core (or, in practice, a paralleled/stacked assembly of standard nanocrystalline C-cores or cut cores), consistent with a 250 kW MF transformer.

Step 3 — Core area and turns

Assuming a roughly square window/core split, $A_c \approx \sqrt{A_p} \approx 37\,\text{cm}^2$. From Eq. (1):

$$N_1 = \frac{V_1}{K_f\, A_c\, B_m\, f} = \frac{800}{4 \times 37 \times 10^{-4} \times 1.0 \times 10{,}000} \approx 5.4 \;\Rightarrow\; N_1 = N_2 = 6 \text{ turns} \tag{8}$$

(rounding up gives margin against saturation; with $n = 1$, $N_2 = N_1$).

Step 4 — Conductor sizing

RMS current per winding (lossless approximation):

$$I_1 = I_2 = \frac{P}{V} = \frac{250{,}000}{800} = 312.5\,\text{A} \tag{9}$$

Required copper cross-section at $J = 300\,\text{A/cm}^2$:

$$A_{Cu} = \frac{I}{J} = \frac{312.5}{300} \approx 1.04\,\text{cm}^2 \tag{10}$$

With skin depth $\delta \approx 0.66\,\text{mm}$ at 10 kHz, this area should be realized as a foil (or paralleled foils) of thickness $\leq 1.3\,\text{mm}$ and sufficient width, or an equivalent litz bundle — foil/busbar is the practical choice at this current level and turn count.

Summary Table

Parameter	Value
Core material	Nanocrystalline (e.g. Vitroperm 500F)
B_m	1.0 T
Area product A_p	$\approx 1389\,\text{cm}^4$
$N_1 = N_2$	6 turns
$I_1 = I_2$ (RMS)	312.5 A
Winding type	Interleaved copper foil, $t \leq 1.3\,\text{mm}$

Notes

This first-pass sizing must be followed by: (i) FEA or analytical leakage inductance estimation (relevant for DAB/PSFB soft-switching range), (ii) thermal verification of core and winding losses against the chosen cooling method, and (iii) insulation coordination for the working voltage and dV/dt. The area-product method gives the correct order of magnitude but final core selection should match an actual manufacturer part (or stack thereof) with $A_p \geq$ the calculated value.